

How to adjust Flash partition

This chapter introduces how to adjust the embedded flash partition and how to modify the partition configuration file.

Adjustment Principle

1. FOTA upgrade or downgrade is not allowed between firmware versions with different partitions.
2. The embedded flash partition adjustment can only be performed among the three partitions, APP, APPIMG and FS. The APP and APPIMG partitions should be 4K aligned, and the FS partition should be 32K aligned.
3. Execute cmd of "build_all.bat new" to compile after partition adjustment.

Configuration File

You can adjust the embedded flash partition of the module based on actual requirements. The embedded flash partition configuration file is located in the directory:

```
components\hal\config\8910
```

Take `partinfo_8910_8m_opencpu.json` as an example, the partition configuration information is shown in the figure below:

```

components > hal > config > 8910 > {} partinfo_8910_8m_opencpu.json > ...
3      "descriptions": [
51      }
52    ],
53    "macros": { Partition address range definition
54      "CONFIG_BOOT_FLASH_ADDRESS": "0x60000000",
55      "CONFIG_BOOT_FLASH_SIZE": "0x10000",
56      "CONFIG_APP_FLASH_ADDRESS": "0x60010000",
57      "CONFIG_APP_FLASH_SIZE": "0x240000",
58      "CONFIG_APPIMG_FLASH_ADDRESS": "0x60250000",
59      "CONFIG_APPIMG_FLASH_SIZE": "0xD0000",
60      "CONFIG_FS_SYS_FLASH_ADDRESS": "0x60320000",
61      "CONFIG_FS_SYS_FLASH_SIZE": "0x180000",
62      "CONFIG_FS_MODEM_FLASH_ADDRESS": "0x604a0000",
63      "CONFIG_FS_MODEM_FLASH_SIZE": "0x340000",
64      "CONFIG_FS_FACTORY_FLASH_ADDRESS": "0x607e0000",
65      "CONFIG_FS_FACTORY_FLASH_SIZE": "0x20000",
66      "CONFIG_FS_SYS_MOUNT_POINT": "/",
67      "CONFIG_FS_MODEM_MOUNT_POINT": "/modem",
68      "CONFIG_FS_FACTORY_MOUNT_POINT": "/factory",
69      "CONFIG_FS_MODEM_NVM_DIR": "/modemnvm",
70      "CONFIG_FS_AP_NVM_DIR": "/nvm",
71      "CONFIG_FS_FOTA_DATA_DIR": "/fota"
72    }

```

Adjustable area

APP
APPIMG
FS

```

components > hal > config > 8910 > {} partinfo_8910_8m_opencpu.json > ...
1    {
2      "version": "0x100", Version number
3      "descriptions": [
4        {
5          "type": "FBD2",
6          "flash": "SFL1",
7          "name": "FSYS",
8          "offset": "0x320000",
9          "size": "0x180000",
10         "erase_block": "0x8000",
11         "logic_block": "0x200"
12       },

```

The description of FS partition information

Adjust size

In the partition configuration file, you can adjust the start address and range among APP, APPIMG and FS only.

1. When adjusting the size of APP and APPIMG, you only need to modify the information of partition address range definition (macros).
2. When adjusting the size of FS partition, you need to modify the information of partition address range definition (macros) and FS partition description (descriptions).

How to adjust RAM partition

We can adjust the RAM size in the file of **target.config**, which is depended on the module type you use. Take an example:

```
components\ql-config\build\EC200UCN_AA\8915DM_cat1_open\target.config
```

Modify the value of **CONFIG_APP_FLASHIMG_RAM_SIZE**. See the figure below:

```
components > ql-config > build > EC200UCN_AA > 8915DM_cat1_open > target.config
1  # Generated by Kconfiglib (https://github.com/ulfalizer/Kconfiglib)
2
3  #
4  # RAM layout
5  #
6  CONFIG_APP_FLASHIMG_RAM_SIZE=0x20000
7
8  CONFIG_CHIP_8910_NOR8M_PSRAM16M=y
9  CONFIG_BOARD_SUPPORT_SIM1_DETECT=y
10 CONFIG_BOARD_SIM1_DETECT_GPIO=4
11 CONFIG_NVM_VARIANTS="$ENV{modemdir}"
12 CONFIG_PARTINFO_JSON_PATH="$ENV{partitionfile}"
13 CONFIG_PINMUX_JSON_PATH="components/hal/config/8910/pinmux_8910_v2.json"
```